

Total No. of Questions : 4]

SEAT No. :

P-5101

[Total No. of Pages : 2

[6187]-508

T.E. (Electronics and Computer Engineering) (In sem)

ADVANCED JAVA PROGRAMMING

(2019 Pattern) (Semester - I) (310342)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) Answer Q1 or Q2, Q3 or Q4
- 2) Figures to the right side indicate full marks.
- 3) Assume suitable data, if necessary.
- 4) Neat diagram must be drawn wherever necessary.

- Q1)** a) Define inheritance in object-oriented programming. Explain the concept of superclass and subclass in inheritance. [5]
- b) Explain the concept of dynamic method dispatch in the context of method overriding. [5]
- c) Compare the access modifiers Public, Protected, Private, Default with respect to access location. [5]

OR

- Q2)** a) Implement a program that showcases multilevel inheritance involving three classes, highlighting the interactions between them. [5]
- b) Provide an example of how to use `super()` to call a constructor in the superclass. [5]
- c) Explain the concept of "volatile" and how it affects class variables in a multithreaded environment. [5]

P.T.O.

- Q3) a) Describe the steps needed to create your own package. Support your answer with an example. [5]
- b) Name some methods in the String class for accessing individual characters. Write Java code to use `getChars()` method. [5]
- c) Write Java code to demonstrate the use of a try-catch block for handling an Arithmetic Exception. [5]

OR

- Q4) a) Explain the working of Java exception hierarchy. Explain the difference between checked and unchecked exceptions. [5]
- b) Enlist some commonly used interfaces in Java's Collections framework? Explain the difference between a Set and a List in Java Collections. [5]
- c) Explain how does a socket facilitate communication between two computers over a network? [5]

Total No. of Questions : 4]

SEAT No. :

P8857

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Oct-22/TE/Insem-608

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Instructions to the candidates :

- 1) Answer Q1 or Q2, Q3 or Q4.
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Q1) a) Describe constructor chaining using this () keyword. [3]

b) Describe different forms of Inheritance with example. [6]

c) Explain Method overriding with suitable example in detail. [6]

OR

Q2) a) Write shorts on Abstract class. [3]

b) Explain in detail about Java Access Modifiers. [6]

c) Explain super keyword with all its usage. Support explanation with an example. [6]

Q3) a) Explain Packages in java with an example. [3]

b) Explain how exceptions can be handled in Java. Give an example using try and catch block. [6]

c) Compose a java socket program that communicates between server and client using TCP sockets. [6]

OR

Q4) a) Describe URL class in java. [3]

b) Explain string handling concept in detail. [6]

c) Write short note on the following : [6]

i) Collections

ii) Maps

